

High-Frequency Identification of Monetary Policy and Information Shocks Across Implied Equity Durations

UNC Quantitative Financial Economics Credential Research Paper

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ABSTRACT

This paper investigates the influence of Federal Open Market Committee (FOMC) announcements on equity prices across varying cash-flow horizons. Existing literature has utilized many high-frequency event studies and Vector Autoregressions (VARs) to document the impact of FOMC communication shocks on derivatives with fixed maturities and bonds, the effects on infinite-horizon equities with varying implied durations remain largely unexplored. To address this gap, we employ a high-frequency framework alongside a measure of implied equity duration to isolate how FOMC shocks, categorized as a pure monetary policy surprise shock and Fed information effect shock, alter risk pricing across the term structure of equity.

We find that the information shock is statistically significant across implied equity durations, whereas the pure monetary policy surprise shock lacks strong statistical significance. Crucially, the information shock disproportionately reprices high-duration equities, emphasizing that when the Federal Reserve reveals private information regarding macroeconomic fundamentals, the market interprets it as a persistent shift in long-term trend growth rather than a temporary cyclical fluctuation. Conversely, the muted response to the monetary policy shock suggests that short-term interest rate surprises, measured via 3-month futures, do not systematically alter the long-run discount rates applied to higher-duration equities.

INTRODUCTION

A substantial body of literature in financial economics seeks to disentangle the effects of Federal Reserve communications on economic outcomes, motivated by the need to understand how the central bank influences markets beyond conventional policy rate adjustments. However, empirical researchers have historically faced a significant methodological hurdle: the endogeneity of monetary policy. Because the Federal Reserve's actions are frequently confounded by contemporaneous financial shocks, isolating the causal impact of its communications is inherently difficult.

To address this identification problem, two primary analytical frameworks have emerged: Structural Vector Autoregressions (SVARs) and high-frequency event studies (Nakamura and Steinsson, 2018). Numerous studies employing these techniques have analyzed how Federal Reserve information affects financial asset outcomes, including Treasury yields, interest rate futures, and equity prices (e.g., Gürkaynak et al., 2005; Bernanke and Kuttner, 2005; Rosa, 2013; Nakamura & Steinsson, 2018; Jarociński & Karadi, 2020; Swanson, 2020; Acosta et al., 2025; Boehm, 2025). These approaches capture the immediate risk adjustment and price discovery of assets during a tight window around an FOMC announcement, allowing a better understanding of FOMC press release impacts.

Based on the foundational framework defined in Gürkaynak et al. (2005) and Bernanke and Kuttner (2005), much of this literature relies on intraday data captured within a narrow, 30-minute window surrounding the announcement to analyze how the FOMC affects equity and bond prices (Swanson, 2021). This tight timeframe is crucial to

the analysis, as it increases the precision of estimates by reducing noise from concurrent macroeconomic shocks, and it prevents the simultaneous equations problem that arises in larger windows, where asset price fluctuations could conversely influence the FOMC's own decisions, and so, their communications to the public on the state of the economy.

Researchers have found factors that affect abnormal returns, such as a high-frequency event study analysis by Gürkaynak et al. (2005), who demonstrated that FOMC statements are not fully captured by a single-factor model relying solely on current federal funds rate target surprises. Instead, asset price responses are better explained by a two-factor model that includes a "future path of policy" factor, capturing forward guidance that moves near-term interest rates without altering the current target. Subsequent literature has built extensively on this multi-factor approach. Swanson (2021) separately identifies surprise changes in the federal funds rate, forward guidance, and large-scale asset purchases, finding that all three components have highly statistically significant effects on Treasury yields, stock prices, and exchange rates. Furthermore, recent work by Boehm and Kroner (2024) uses a heteroskedasticity identification method using a Rigobon estimator to confirm that the current federal funds rate target surprise accounts for only a small fraction of the variation in asset prices and dollar exchange rates, reaffirming the necessity of accounting for broader FOMC communication shocks.

Using this framework, our paper uses two foundational shocks to model FOMC announcements: the pure monetary policy surprise shock and the Federal Reserve information effect shock, created by Jarociński and Karadi (2020). As Jarociński and

Karadi (2020) demonstrate, central bank announcements simultaneously convey unexpected variation in policy alongside the central bank's proprietary assessment of the economic outlook. The pure monetary policy shock behaves according to standard macroeconomic theory: an unexpected policy tightening raises the discount rate and reduces expected future dividends due to a deteriorating outlook, causing stock prices to fall. This creates a strict negative co-movement between interest rates and equity valuations. Conversely, the information effect describes the process by which the private sector updates its beliefs about underlying macroeconomic fundamentals based on central bank actions and communications (Nakamura & Steinsson, 2018). Under this dynamic, a contractionary policy announcement can simultaneously reveal positive private information about the economy's current strength. If the resulting upward revision in expected future dividends outweighs the negative impact of a higher discount rate, it leads to a positive co-movement where both interest rates and stock prices rise. Consequently, contractionary announcements can sometimes actually raise expectations for output growth, and so stock prices as well. While standard high-frequency event studies often conflate these signals since they are concurrent, which leads to heavily biased estimates of monetary nonneutrality, Jarociński and Karadi (2020) successfully isolate these two distinct shocks utilizing a structural vector autoregression (SVAR) framework. By imposing strict sign restrictions on the intraday co-movement of interest rates and stock prices, their methodology allows them to isolate the monetary policy shock from the confounding noise of the information effect.

Building upon the theoretical debates about the strength of these factor effects established by Ai and Hu (2025) and Bauer and Swanson (2023), this paper seeks to

quantify the relative strength of the pure monetary policy shock versus the information shock. We advance this framework by explicitly exploring how the intensity of these dual shocks varies across different durations of equities.

Multiple studies have provided the foundation for analyzing the strengths of the monetary policy and information shocks. For monetary policy surprise shocks, Bernanke and Kuttner (2005), concluded that most of the stock market's reaction to monetary policy surprises is driven by changes in expected excess returns, pointing to a significant impact on the equity risk premium. However, recent literature has challenged this view. Nagel and Xu (2024) demonstrate that the equity market's response to FOMC announcements is almost entirely explained by changes in the default-free term structure of yields, a pure discount rate effect, rather than shifts in the equity premium. Parallel to this debate over pure monetary shocks, the magnitude of the information shock is also subject to varying structural conditions. As Ai and Hu (2025) note, the Fed information effect is not static; its market impact is strongest when uncertainty regarding the central bank's long-run policy objectives is elevated. Ultimately, the ongoing literature underscores that the overall strength and transmission of Federal Reserve communication shocks depend heavily on both the underlying economic environment and the specific nature of the asset being priced.

Many of these studies rely on blanket market risk premiums, and less on different durations of equities. However, a growing subfield within this literature seeks to analyze the term structure of equity with regards to these communications shocks. For instance, Hamoud (2025), measured these distinct cash-flow horizons by relying on the fixed contractual maturities of derivative markets, and, using the method of dividend strips

from van Binsbergen & Koijen, (2017) uses daily regressions of dividend futures to trace the impact of monetary and information shocks across specific contractual maturity brackets. Specifically, Hamoud successfully traces the impact of FOMC announcements across different contractual maturities, isolating how these dual factors impact specific equity derivatives. However, relying on daily data reintroduces the potential for macroeconomic noise and endogeneity that high-frequency identification is specifically designed to eliminate (Gürkaynak et al., 2005). Furthermore, analyzing fixed contractual horizons limits the applicability of these findings to the broader, underlying equity market that does not typically have contractual maturities.

To address these limitations, our paper turns to an alternative measure of the equity term structure: implied equity duration (Dechow et al., 2002; Weber, 2018). This analysis extends Hamoud's foundational concept but shifts the focus to the broader market by utilizing Jarociński and Karadi's (2020) calculated monetary policy and information shocks on infinite duration equities using a high-frequency event study analysis and a firm fixed effects model. Ultimately, this approach allows us to strictly identify the differential effects of Federal Reserve communications on equities with systematically distinct, infinite cash-flow horizons.

Dechow et al. (2002) provides the foundational framework for this approach by adapting Macaulay's traditional fixed-income duration formula to equity markets to calculate the implied duration of equities. Since equities lack a fixed maturity date, their formulation partitions the timeline into a finite forecasting horizon based on historical financial data and an infinite cash flow horizon calculated as a perpetuity. Dechow et

al.'s approach to calculating implied equity duration has been subsequently used for a variety of applications, ensuring its credibility as a measure of underlying equity risk.

Weber (2018) utilizes autoregressive processes to model equity returns and growth, demonstrating that implied equity duration effectively captures cross-sectional risk characteristics. Specifically, Weber finds that equity betas, overall stock return volatility, and idiosyncratic volatility all scale positively with duration. Furthermore, he identifies a downward-sloping term structure of equity returns, noting that short-duration assets paying near-term dividends historically outperform high-duration stocks, largely due to downward adjustments on overly optimistic analyst projections for distant cash flows.

More recently, Dechow et al. (2021) use implied equity duration to analyze the sensitivity of equity prices to pandemic shutdowns, noting that short-term cash flows bore the brunt of the shock, meaning low-duration equities experienced greater price disruptions. Collectively, these findings validate implied equity duration as a rigorous metric for mapping the term structure of the broader stock market.

By integrating this measure into our high-frequency event study framework and Jarociński and Karadi's (2020) shocks, we can reliably test how the competing forces of pure monetary policy and central bank information shocks systematically reprice risk across distinct implied equity cash-flow horizons.

DATA DESCRIPTION

We base our methodology on the implied equity duration framework established by Dechow et al. (2004), and subsequently utilized by Weber (2018) and Dechow et al. (2021). To construct our dataset, we utilize the similar datasets employed by Weber (2018) and Dechow et al. (2004, 2021): We source daily stock return data for publicly listed firms in North America from the CRSP over the period 2000–2024. Market excess returns and the risk-free rate are obtained from the Kenneth French Data Library. We additionally obtain quarterly firm-level accounting data from Compustat Fundamentals Annual through WRDS over fiscal years 2000–2024, including sales, income before extraordinary items, shareholders' equity, fiscal year-end prices, and shares outstanding. Market capitalization is constructed as the product of price and shares outstanding.

Following the Dechow et al. (2004) framework, we estimate future cash flows across two distinct periods: a finite forecasting horizon denoted as T , and an infinite terminal perpetuity representing all cash flows beyond this finite horizon.

The core calculation adapts the traditional Macaulay duration formula for fixed-income assets by substituting fixed coupon payments with expected equity cash flows. The duration equation relies on the current market value, denoted as $P_{i,t}$, the discount rate, denoted as r , and the expected cash flow of firm i at time $t + s$, denoted as $CF_{i,t+s}$. The resulting formula weights the time period of each cash flow by its present value relative to the total price of the asset. Because the asset has infinite cash flows, the second term of the equation calculates the duration of a perpetuity, which is

mathematically defined as $T + (1 + r)/r$, and weights it by the terminal value of the equity.

To forecast the cash flows, we rely on the clean surplus accounting identity, defined by Weber (2018), which dictates that cash flows are equivalent to earnings minus the change in book equity. Under this assumption, cash flows are derived by forecasting the return on equity, denoted as $ROE_{i,t+s}$, and the growth in book equity, denoted as $g_{i,t+s}$.

The return on equity is defined as the earnings of the firm at time $t + s$, divided by the book value of the firm's equity at the prior period, denoted as $BV_{i,t+s-1}$ (Dechow et al., (2004). The growth in book equity is defined as the change in book value from period $t + s - 1$ to period $t + s$, divided by the initial book value $BV_{i,t+s-1}$ (Dechow et al., (2004). By substituting these definitions into the clean surplus relation, the future cash flows of the asset can be shown as a function of the book value, return on equity, and growth rate:

$$CF_{i,t+s} = BV_{i,t+s-1}(ROE_{i,t+s} - g_{i,t+s})$$

Then, the formula for implied equity duration is:

$$D_{i,t} = \frac{\sum_{s=1}^T s \cdot \frac{CF_{i,t+s}}{(1+r)^s}}{P_{i,t}} + \left(T + \frac{1+r}{r}\right) \cdot \frac{P_{i,t} - \sum_{s=1}^T \frac{CF_{i,t+s}}{(1+r)^s}}{P_{i,t}}$$

To accurately forecast these components, we follow the conventions in Dechow (2002), Dechow (2021), and Weber (2018). We model the forecasted variables as

autoregressive processes and assume that the return on equity and sales growth are mean-reverting processes. We follow the latest research from Dechow (2021), and adopt the first-order autoregressive process with coefficient of 0.39 and the mean of the discount rate of 0.12 for return on equity; the first order autoregressive process with coefficient of 0.21 and growth rate of 0.06 for book equity. These choices are consistent with previous studies. Finally, we utilize a finite forecasting period, T , of 15 years before the terminal perpetuity assumption takes effect.

Following the theoretical framework of Jarociński and Karadi (2020), each FOMC communication is assumed to generate two distinct types of shocks: a monetary policy surprise shock and a central bank information shock. The goal of analyzing high-frequency financial market movements around FOMC announcements is to identify and quantify these latent shocks. We obtain the FOMC market shock data from Marek Jarociński's github page. The data provides the monetary policy surprise shock and the information shock for each FOMC event during the time frame February 1988 - January 2024 (Jarociński and Karadi, 2020).

Our panel event study framework computes cumulative abnormal returns (CAR) around FOMC communication events. To implement this, we require daily equity prices, market returns, and the risk-free rate. We estimate expected equity returns using the CAPM, based on information available prior to the FOMC announcement, and use these estimates to construct abnormal returns in the post-event window. Equity prices are obtained from CRSP Annual accounting (fundamental) data for. The market return data are from daily value-weighted returns on all CRSP common stocks, including

dividends. Risk-free rates are sourced from the Fama–French Daily Research Factors dataset.

Empirical Methods

Our empirical strategy consists of two steps. First, we use the method in the Data Description section to calculate the implied duration for each firm prior to each FOMC event date. For example, for a FOMC event that happened on 2001-03-20, we associate the implied duration of 2001 February with it. In this case, applying the March implied duration leads to look ahead bias, because the March implied duration contains information after the FOMC event.

Second, we estimate a panel-based event study model using these identified shocks. We propose a panel event study model to examine how equity returns respond to monetary policy shock and information shock vary with firm's implied duration. The second step of our method starts with estimating firm specific returns using the CAPM. For each FOMC communication data and each firm, we use the interval between the previous and current FOMC meetings as the estimation window to construct the following model:

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_i (R_{m,t} - R_{f,t}) + \epsilon_{i,t}$$

where $R_{i,t} = \ln(P_{i,n+}) - \ln(P_{i,n-})$ is the log difference return calculated by log today's price subtracts log yesterday's price. $R_{m,t}$ is the market return and $R_{f,t}$ is the risk-free rate.

Using the estimated parameters, we forecast expected returns over 5 days after the event and constant abnormal return as the difference between realized and expected returns. These abnormal returns are then aggregated into cumulative abnormal returns (CAR) over the event window, providing a firm-level measure of the market reaction to each FOMC announcement.

We then link these CAR to cross-sectional variation in implied duration. For each FOMC event, firms are sorted into 20 quantiles based on their pre-event implied duration, measured using the most recent available information prior to the announcement. This procedure assigns firms into groups, where higher quantiles correspond to firms with longer implied duration. $D_{i,t-}^{(q)}$ is the dummy variable:

$$D_{i,t-}^{(q)} = \begin{cases} 1 & \text{if } D_{i,t-} \in Q_q, \\ 0 & \text{otherwise,} \end{cases}$$

This means if the implied duration of a firm at a specific event date falls into the q-th quantile of the list of sorted implied duration, the dummy is 1, otherwise the dummy is 0. This design allows us to estimate 38 coefficients in the regression.

Specifically, we estimate the panel model with the firm fixed effects α_i . To allow for heterogeneous responses to monetary policy and information shocks, we interact the shocks with duration dummies, and the lowest-duration group is the reference group. Instead of event fixed effects, we include the raw monetary policy and information shocks directly, which identify the baseline response of the reference group.

$$CAR_{i,t} = \delta^{MP} \varepsilon_t^{MP} + \delta^{INFO} \varepsilon_t^{INFO} + \sum_{q=2}^{20} \lambda_q \left(D_{i,t-}^{(q)} \varepsilon_t^{MP} \right) + \sum_{q=2}^{20} \gamma_q \left(D_{i,t-}^{(q)} \varepsilon_t^{INFO} \right) + \theta \left(\beta_i^{CAPM} \varepsilon_t^{MP} \right) + \phi \left(\beta_i^{CAPM} \varepsilon_t^{INFO} \right) + \alpha_i + u_{i,t}$$

The $D_{i,t-}^{(q)} \varepsilon_t$ terms measure how the CAR of equities in q-th implied duration quantile respond to FOMC shocks. We expect that the values of λ_q and γ_q are significantly different with the variation of q . By the model specifications, $q = 1$ is the reference group, since its implied duration dummy is forced to be 0. δ^{MP} and δ^{INFO} terms are the baseline slope coefficients capturing the effect of shocks on the reference group. Thus, the economic interpretation of estimated λ_q and γ_q is the differential response of higher-implied duration equities relative to the lowest-implied duration equities. ε_t^{MP} and ε_t^{INFO} are FOMC market policy and information shocks in each event. We construct interaction terms by combining them with β_i^{CAPM} , the same market risk factor in the CAPM estimation.

To ensure the results are driven by cash-flow horizons rather than systematic risk exposures, we included an interaction term between the firm's CAPM beta and the respective FOMC shocks. Because equity duration and market beta are inherently

correlated, omitting this interaction would conflate the cash-flow timing effect with broader market risk premium differences. By controlling for systematic risk in this way, we are able to distinguish whether the cross-sectional variation in responses to FOMC shocks reflects differences in implied duration or differences in systematic risk exposure, isolating the pure duration effect and ensuring that any observed variation is attributable to differences in cash-flow horizons rather than market risk factors.

Regression Results

We run the panel model in 20 quantiles, and obtained coefficients for MP and INFO shocks. The following regression summary table presents the regression results. Almost all coefficients are significant at 1% level, with a few exceptions are significant at 10% level.

Panel A: Overall Sample Summary

Metric	Value
Observations	965,551
Firms	13,890
Events	193
Dep. var mean	-0.0017
Dep. var std	0.1141
Duration mean	16.4523
Duration median	18.8301
R ²	0.0008
Adj. R ²	0.0008
Covariance	HC1

Model
Reference bin

binned_withbeta
1

Panel B: Coefficient Estimates

Duration portfolio	Coef	Std Err	z	P> z
eps_MP	-0.1523	0.0196	-7.7605	0.0000
eps_INFO	-0.2342	0.0439	-5.3373	0.0000
MP_q2	0.0673	0.0241	2.7884	0.0053
MP_q3	0.0570	0.0227	2.5112	0.0120
MP_q4	0.0923	0.0228	4.0463	0.0001
MP_q5	0.0804	0.0216	3.7164	0.0002
MP_q6	0.0604	0.0213	2.8333	0.0046
MP_q7	0.0589	0.0214	2.7566	0.0058
MP_q8	0.0418	0.0220	1.9060	0.0566
MP_q9	0.0685	0.0221	3.0916	0.0020
MP_q10	0.0622	0.0217	2.8661	0.0042
MP_q11	0.0499	0.0215	2.3253	0.0201
MP_q12	0.0390	0.0215	1.8165	0.0693
MP_q13	0.0506	0.0217	2.3378	0.0194
MP_q14	0.0658	0.0218	3.0175	0.0025
MP_q15	0.0770	0.0219	3.5240	0.0004
MP_q16	0.0551	0.0218	2.5236	0.0116
MP_q17	0.0816	0.0232	3.5255	0.0004
MP_q18	0.0537	0.0227	2.3695	0.0178
MP_q19	0.0560	0.0243	2.3050	0.0212
MP_q20	0.0873	0.0273	3.1978	0.0014
INFO_q2	0.0942	0.0550	1.7134	0.0866
INFO_q3	0.1688	0.0515	3.2775	0.0010
INFO_q4	0.1762	0.0513	3.4332	0.0006
INFO_q5	0.1644	0.0537	3.0596	0.0022

INFO_q6	0.2312	0.0483	4.7911	0.0000
INFO_q7	0.2463	0.0479	5.1465	0.0000
INFO_q8	0.2381	0.0487	4.8853	0.0000
INFO_q9	0.2535	0.0485	5.2220	0.0000
INFO_q10	0.2542	0.0488	5.2083	0.0000
INFO_q11	0.2788	0.0494	5.6482	0.0000
INFO_q12	0.2421	0.0512	4.7327	0.0000
INFO_q13	0.2479	0.0498	4.9770	0.0000
INFO_q14	0.2561	0.0498	5.1384	0.0000
INFO_q15	0.2655	0.0522	5.0830	0.0000
INFO_q16	0.3025	0.0517	5.8455	0.0000
INFO_q17	0.2636	0.0553	4.7682	0.0000
INFO_q18	0.3156	0.0580	5.4420	0.0000
INFO_q19	0.2660	0.0629	4.2281	0.0000
INFO_q20	0.2184	0.0703	3.1050	0.0019
DELTA_MP	0.0965	0.0071	13.5776	0.0000
DELTA_INFO	-0.0738	0.0169	-4.3675	0.0000

Notes: Panel A reports overall sample and model summary statistics. Panel B reports coefficient estimates in a statsmodels-style block for direct pasting into Google Docs.

Figure 1 demonstrates how MP and INFO shock exposure varies across implied duration quantiles. Visually, INFO shock coefficient increases significantly when $q < 7$, then it becomes relatively stable. However, the variation of MP shock coefficient is not obvious across implied duration.

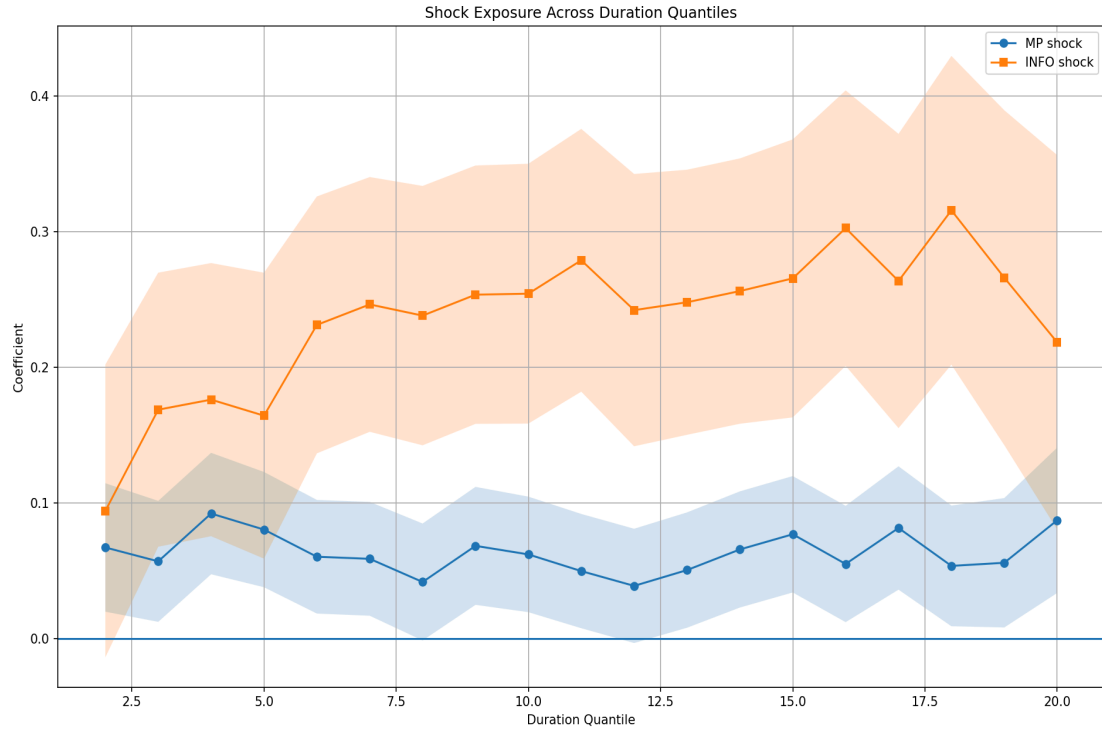


Figure 1: Shock Exposure across Duration Quantiles

We run an OLS regression on them to test if their estimated slopes are positive. The following table presents the result summary. We reject the null at 1% significance level for INFO shock coefficients, but we cannot reject the null for MP shock coefficients.

Monotonicity Test of Duration-Bin Coefficients

Weighted least squares trend regressions by shock type

Shock	Bins	Slope on Bin	Std. Err.	t-stat	p-value (2-sided)	p-value (1-sided, increasing)	R ²	Intercept
Monetary Policy	19	-0.0002	(0.0007)	-0.2783	0.7841	0.6079	0.0045	0.0647
Information	19	0.0074***	(0.0015)	4.8543	0.0001	0.0001	0.5809	0.1591

Notes: Each row reports a weighted least squares regression of coefficient estimates on duration-bin index, using inverse-variance weights $1 / SE^2$. “Slope on bin” is the estimated linear trend in coefficients across bins. Standard errors are shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The robust significance of the information shock across quantiles suggests that when the Federal Reserve reveals private information regarding macroeconomic fundamentals, it triggers a heterogeneous revision of expected future cash flows across the equity term structure. Specifically, our results indicate that the information shock disproportionately reprices high-duration assets compared to their short-duration counterparts. Theoretically, this suggests that the market interprets the central bank's information not merely as a signal of temporary cyclical fluctuations, but as an indicator of persistent macroeconomic conditions and shifts in long-term trend growth. Because the valuations of high-duration equities are anchored by distant cash flows and infinite terminal perpetuities, revisions to long-term macroeconomic expectations compound over time, demanding a significantly larger price correction than for assets whose cash flows are concentrated in the near term.

Conversely, the relative insignificance of the MP shock across varying durations highlights the disconnect between short-term policy instruments and long-term equity valuations. Mathematically, duration measures price sensitivity to discount rates; therefore, standard asset pricing theory dictates that a pure monetary policy surprise should disproportionately impact high-duration equities. However, because our MP shock is measured via 3-month futures, it inherently captures only a short-term interest rate surprise. High-duration equities, by contrast, are primarily sensitive to long-term discount rates. Consequently, our findings suggest that a surprise in the immediate fed funds target does not systematically transmit to the long end of the yield curve, thereby failing to uniformly reprice infinite-horizon equities.

CONCLUSION

This paper investigates how two FOMC communication shocks, the pure monetary policy shock and central bank information shock, transmit across the term structure of equity. Leveraging the identification strategy of Jarociński and Karadi (2020) alongside implied equity duration, we provide novel evidence that the equity market's response to Federal Reserve communications is highly heterogeneous depending on cash-flow horizons.

Specifically, we find a robust, statistically significant relationship between the central bank information shock and cumulative abnormal returns across varying equity durations. Crucially, our results indicate that this shock disproportionately reprices high-duration assets compared to their short-duration counterparts. This finding supports the theoretical premise that when the Federal Reserve reveals their own private information on macroeconomic fundamentals, the market interprets it as an indicator as a persistent shift in long-term trend growth. Because the valuations of high-duration equities are heavily influenced by distant cash flows and infinite terminal perpetuities, revisions to long-term macroeconomic expectations compound over time, driving a significantly larger price correction. Conversely, the pure monetary policy shock did not exhibit the same robust cross-sectional variation. This muted response suggests that short-term policy rate surprises, captured via 3-month futures, do not systematically alter the long-run discount rates applied to infinite-horizon equities.

We acknowledge certain data constraints within our current methodology that warrant further investigation. While the FOMC shocks are cleanly identified within a precise 30-minute high-frequency window, our cumulative abnormal returns (CAR) are estimated over a 5-day post-event window due to limitations in intraday equity data

availability. Consequently, our estimates may absorb some degree of concurrent macroeconomic noise. Future research should utilize intraday data to calculate pure high-frequency equity returns, which would more cleanly disentangle these dual shocks from broader market movements.

Nonetheless, our findings offer compelling evidence that the term structure of equity is a vital channel for understanding the transmission of central bank information. Resolving the duration puzzle surrounding pure monetary policy shocks, ideally using strictly high-frequency intraday data, will continue to refine our understanding of how central bank communications steer financial markets.

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